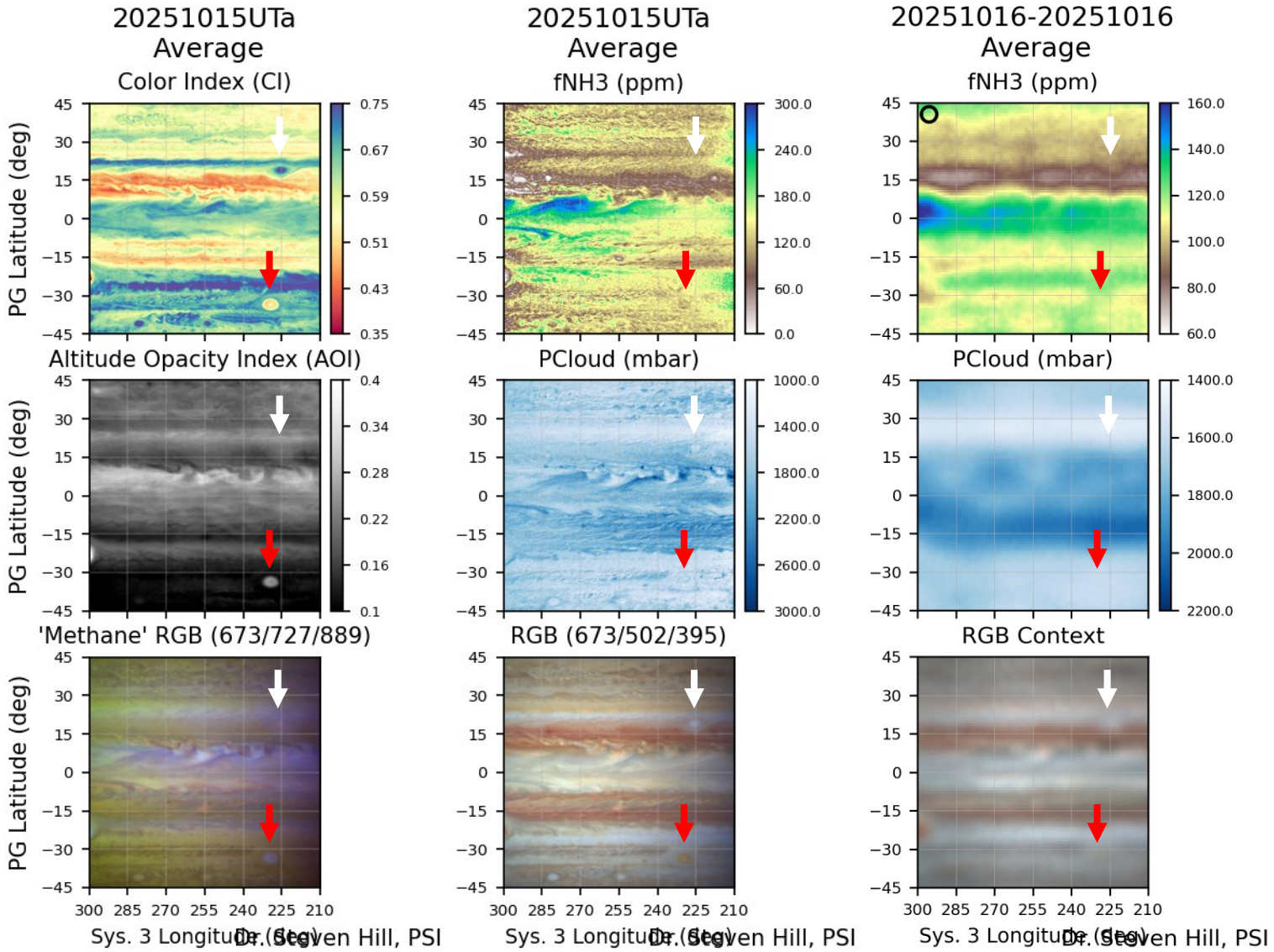
Some Initial Results

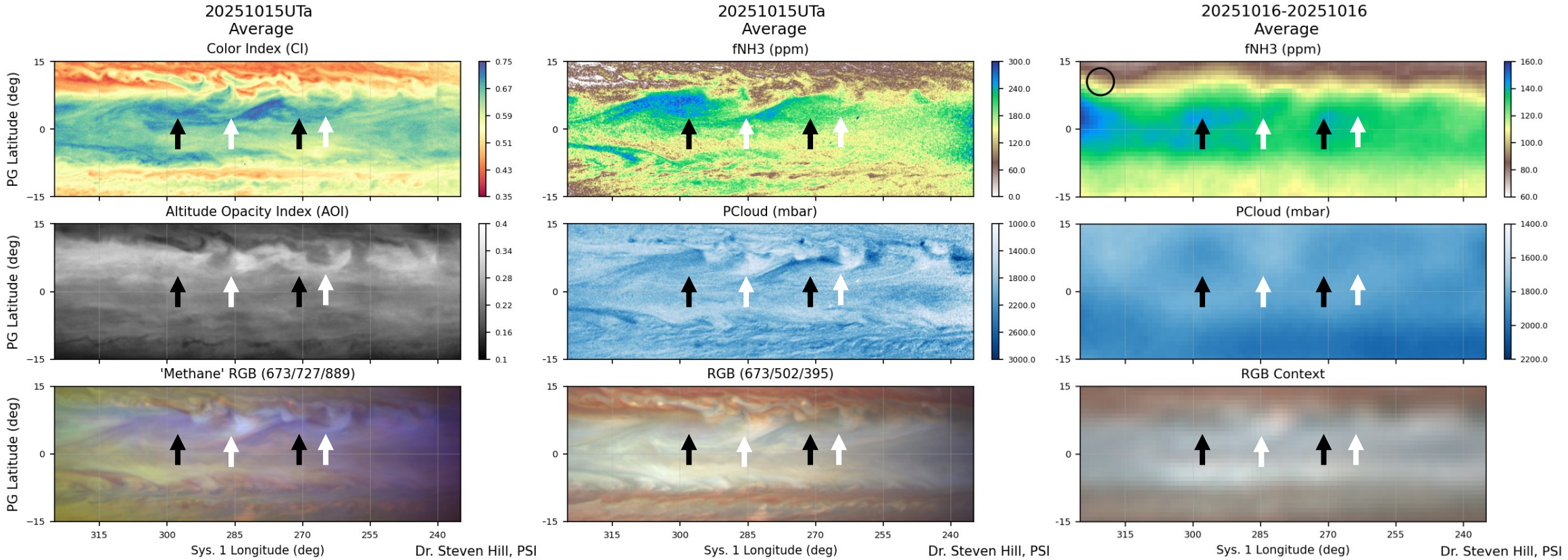
- Data from October 15-16 and November 20, 2025 have been processed from the maps Mike Wong generated
  - Navigation is very good (ratios needed for f_{NH3} and P_{Cld} are sensitive to misalignments)
  - The November data have a ~16-minute gap between the 619 nm exposure and the 631 and 645 nm exposures exacerbating viewing angle corrections needed
  - For now, viewing angle correction is done for both October and November at the data-level (not radiance-level) and is currently purely empirical
  - Flattening is done by using a 2nd order polynomial fit to large-scale longitude variation
- AOI, CI, f_{NH3}, and P_{Cld} are computed along with two RGB composites
  - AOI=$(I/F)_{889}/(I/F)_{275}$ and CI= $(I/F)_{395}/(I/F)_{631}$
  - f_{NH3} and P_{Cld} per Hill *et al.* (2024) and Irwin *et al.* (2025)
  - RGB(visual)=(673,502,395), RGB(NIR)=(673,727,889)
- Absorption coefficients computed for HST
  - K_eff_NH3_645 = 3.090 and K_eff_CH4_619 = 0.548
  - HST f_{NH3} and P_{Cld} show greater extremes than ground-based due to higher resolution

HST Data (Sys 3) from October

- Random Observations:
  - EZ shows greatest variation in P_{Cld} and f_{NH3}, but neither AOI nor CI appears to correlate broadly with P_{Cld}
  - White oval at 10°N (white arrow) shows slightly enriched f_{NH3} and lower P_{Cld} but barely noticeable in AOI
  - AOI shows NTrZ-NTB region (17°-31°N) is multi-hued, but P_{Cld} and f_{NH3} variations are far less dramatic and poorly correlated
  - Red oval at 33°S (red arrow) is prominent in AOI, but not in P_{Cld}, and shows slight f_{NH3} depletion

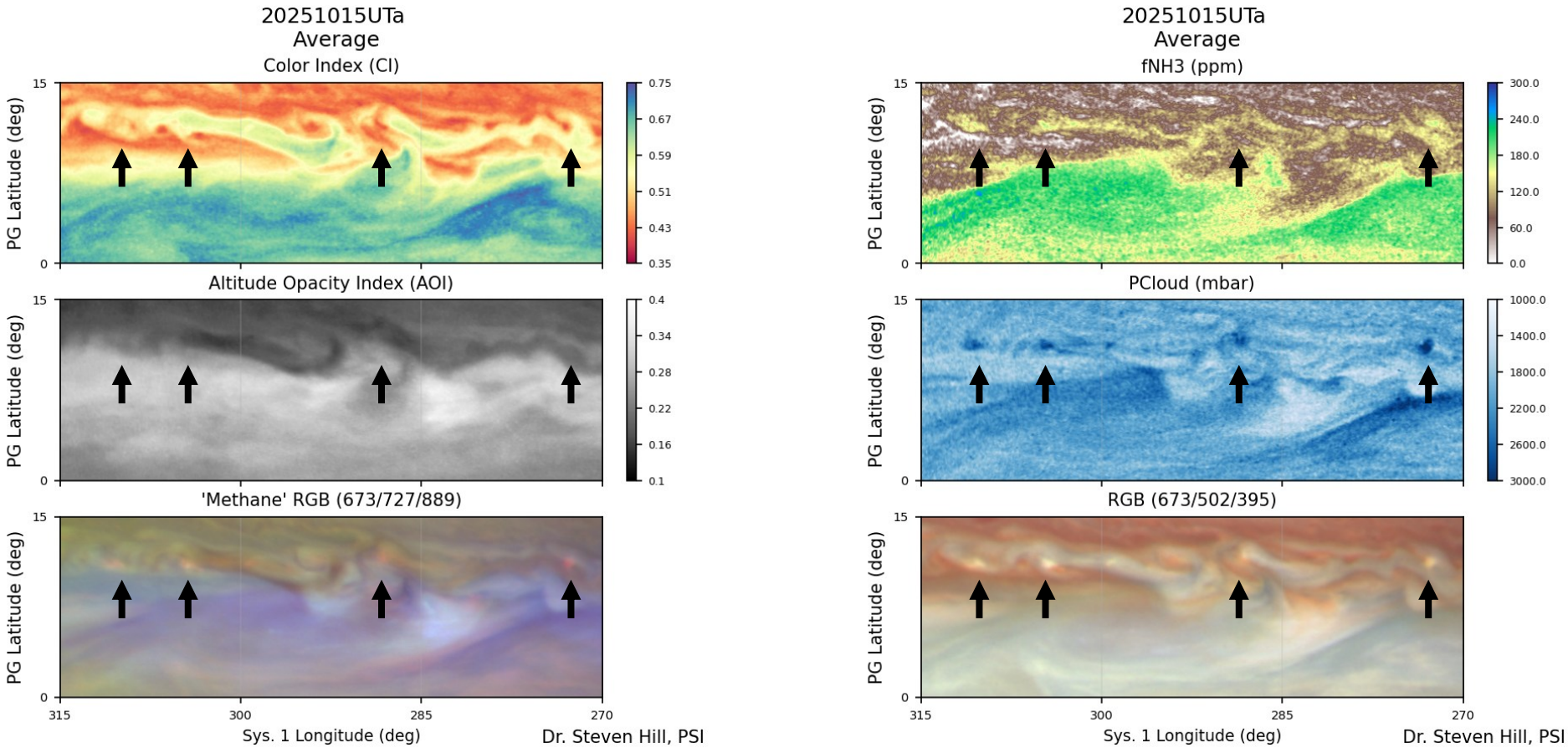


HST Data (Sys 1) from October



- High altitude 'bow' (outflow?) clouds (white arrows) show in P_{Cld} despite the fact it should be probing ~2 bar
- Hotspots (black arrows) have a complex relationship with f_{NH3}

HST Data (Sys 1) from October



- Deep clouds (red in the 'Methane RGB', lower left) show up clearly in the P_{Cld} data as dark spots (high pressure)
- No clear pattern is seen regarding f_{NH3} for these deep clouds